

Aidan Hirsch

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EDUCATION

Purdue University

Bachelor of Science in Biomedical Engineering

3.53/4.0 GPA – May 2028

West Lafayette, IN

RESEARCH EXPERIENCE

Research Assistant

Dadarlat Lab – Purdue University

Jan 2026 – Present

West Lafayette, IN

- Modeled and labeled forelimb movements in mouse behavioral videos for DeepLabCut pose estimation model to be used in conjunction with two-photon microscopy neural data for motor cortex analysis
- Rebuilt and maintained lab's DeepLabCut training and analysis pipeline, restoring video pose estimation for multiple active experiments
- Trained electrode-implanted mice on somatosensory cortex behavioral tasks for neural stimulation studies

Research Assistant

Neuroprostheses Research Lab – Purdue University

Jan 2025 – Dec 2025

West Lafayette, IN

- Discovered novel correlation between modeled circuit values and electrode degradation by analyzing in-vivo electrical component parameters through UMAP statistical analysis
- Analyzed ultramicroelectrode array failure modes using 16-week electrochemical impedance spectroscopy study in rat somatosensory cortex
- Engineered reusable automation pipeline for data pre-processing, measurement modeling, and file preparation, actively in use and saving over 9 months of manual labor and enabling rapid model iteration

Research Assistant

Friel Lab – Burke Neurological Institute

Apr 2021 – May 2024

White Plains, NY

- Designed quantitative bimanual hand assessment for children with cerebral palsy, boosting joint location accuracy by 30-50% over prior qualitative methods
- Engineered a custom drawer for rehabilitation hand assessment using Raspberry Pi and 3D cameras to capture participant joint locations in 3D space
- Developed Python program to interact with drawer assessment by determining when a participant moved or touched parts of the drawer through measurement of electrostatic discharge

LEADERSHIP AND PROJECTS

Chair

Purdue IEEE Systems Man and Cybernetics Committee

Jan 2026 – Present

West Lafayette, IN

- Leading Purdue's only neurotech club with 55+ active members in development of 128-channel EEG, concussion detection scoping research and publications, and international neural decoding competitions
- Teaching various aspects of BCI technology, including ICMS, microelectrode arrays, neurotech market developments, neural anatomy, and computational neuroscience on a weekly basis to club members
- Spearheading PCB design workshops for BME and ECE undergraduates to learn how to design PCBs from schematic, to layout, to part selection, to completed board

EEG Build Team Lead

Purdue IEEE Systems Man and Cybernetics Committee

Aug 2025 – Present

West Lafayette, IN

- Directing Electrical team in engineering a custom 128-electrode EEG system from component level encompassing custom ADC and microcontroller board layouts
- Designed, routed, and fabricated multi-layer printed circuit boards to integrate 16 ADS1299 ADCs to 5 ESP32-S3 microcontrollers to achieve high SNR neural signal acquisition
- Implementing embedded systems network for ESP microcontrollers to synchronize and stream data from all 128 electrode channels

SKILLS

Engineering Tools: KiCad | LTSpice | DeepLabCut | Solidworks | Fusion

Data Analysis: Python (pandas, NumPy, matplotlib, seaborn) | CEBRA | MATLAB | R | C

Biomedical Techniques: ICMS | Impedance spectroscopy | Rodent handling (IACUC) | Cyclic Voltammetry